Data Management (A.A. 2025/26) – exam of 21/09/2026

Problem 1. Consider the two transactions $T_1 = r_1(A)\, r_1(B)\, w_1(B)\, w_1(A)$, and $T_2 = r_2(A)\, r_2(B)\, w_2(B)\, w_2(A)$, and tell how many conflict-serializable schedules on the set $\{T_1, T_2\}$ of transactions exist, motivating the answer in detail.

Problem 2. Provide an example of three transactions, which have the following properties:

(i) When being executed with a 2PL scheduler, deadlock may occur.
(ii) For each pair of the three transactions, no deadlock can occur in any execution of that pair with a 2PL scheduler.

Motivate your answer by providing the definition of 2PL scheduler and proving that the three proposed transactions satisfy both properties *(i)* and *(ii)*.

Problem 3. Let R be a relation stored in a heap file consisting of 10,000 pages. Assume that 20 buffer frames are available.

1. Sort R using external sorting, describe its execution, and compute its cost in terms of page accesses.
2. What is the minimum number of buffer frames needed to sort R with this algorithm in only two passes?
3. Assume now the same setting as in Point 1. However, due to system overload, after the initial runs have been written to secondary storage, the number of available buffer frames is reduced to one quarter of the original number (i.e., after Pass 0, only 5 buffer frames are available). Describe the execution of the external sorting algorithm in this case and compute its cost in terms of page accesses.

Problem 4. Let $R(A, B)$ and $S(B, C)$ be two relations with 500 and 20,000 pages, respectively. Assume that 50 buffer frames are available.

Consider the computation of the natural join $R \bowtie S$ on the common attribute B.

Discuss the possible strategies for computing the join, considering two-pass and multi-pass hash-based and sort-based algorithms. For each of the two families of algorithms (hash-based and sort-based), identify the most efficient applicable strategy and compute its cost in terms of page accesses. Clearly state any assumptions on the data that are required for the proposed algorithms to be applicable. Assume that **both relations are initially stored in heap files and that no indexes are available**.

Problem 5. (only for students who opted for **option 1**, i.e., who do **not** do the project)

1. Create a property-graph database representing the following information (you may draw a labeled graph with properties specified between braces, as in the graphical representation of Neo4j graphs).
   - A person whose name is Roberto teaches a course called Databases in 2026.
   - A professor whose name is Maurizio teaches a course called Databases in 2024 and 2025.
   - Maurizio has been a member of DIAG since 2005.
   - DIAG is a department, founded in 2005.
   - Maurizio and Roberto are Lecturers.
2. Write an RDF dataset G to express the information described at Point 1, but replace the last point with "everyone who teaches is a Lecturer". Use the RDF/RDFS vocabulary where appropriate; otherwise, use any URI you prefer (assuming an empty prefix).
3. Write the SPARQL query Q that retrieves all Lecturers.
4. Provide the answers to Q when it is (a) evaluated over G interpreted as a graph database; and (b) computed over G interpreted as an RDFS Knowledge Graph (i.e., considering also the facts logically implied by G).

Justify your answers.